

MIE366 – DESIGN ASSIGNMENT 1: “DESIGN PROPOSAL”

Department of Mechanical and Industrial Engineering – University of Toronto

[DUE DATE: September 21st, end-of-day, Toronto time]

[GROUP SIZE: 2]

[WEIGHT: 4%]

[ONLINE SUBMISSION]

If you have not yet read the “background” document, please do so now. In this assignment, you are to design the circuit that occupies the ‘CURRENT SENSE’ block in Figure 1 of that document.

CURRENT SENSE DESIGN

Examining Figure 1 from the “background” document, we can see the circuit block dedicated to this assignment marked in green. This “Current Sense Amplifier” measures the current flowing through the load and converts it to a proportional voltage signal. In general current sensing can be achieved in a number of ways, but we have chosen the most direct – we insert a small resistor in series with the load. When the load draws current, a voltage drop develops across this resistor which we can measure, and which will be proportional to the load current according to $V_{SENSE} = I_{LOAD} R_{SENSE}$, where I_{LOAD} is the load current and R_{SENSE} is the sense resistor value used:

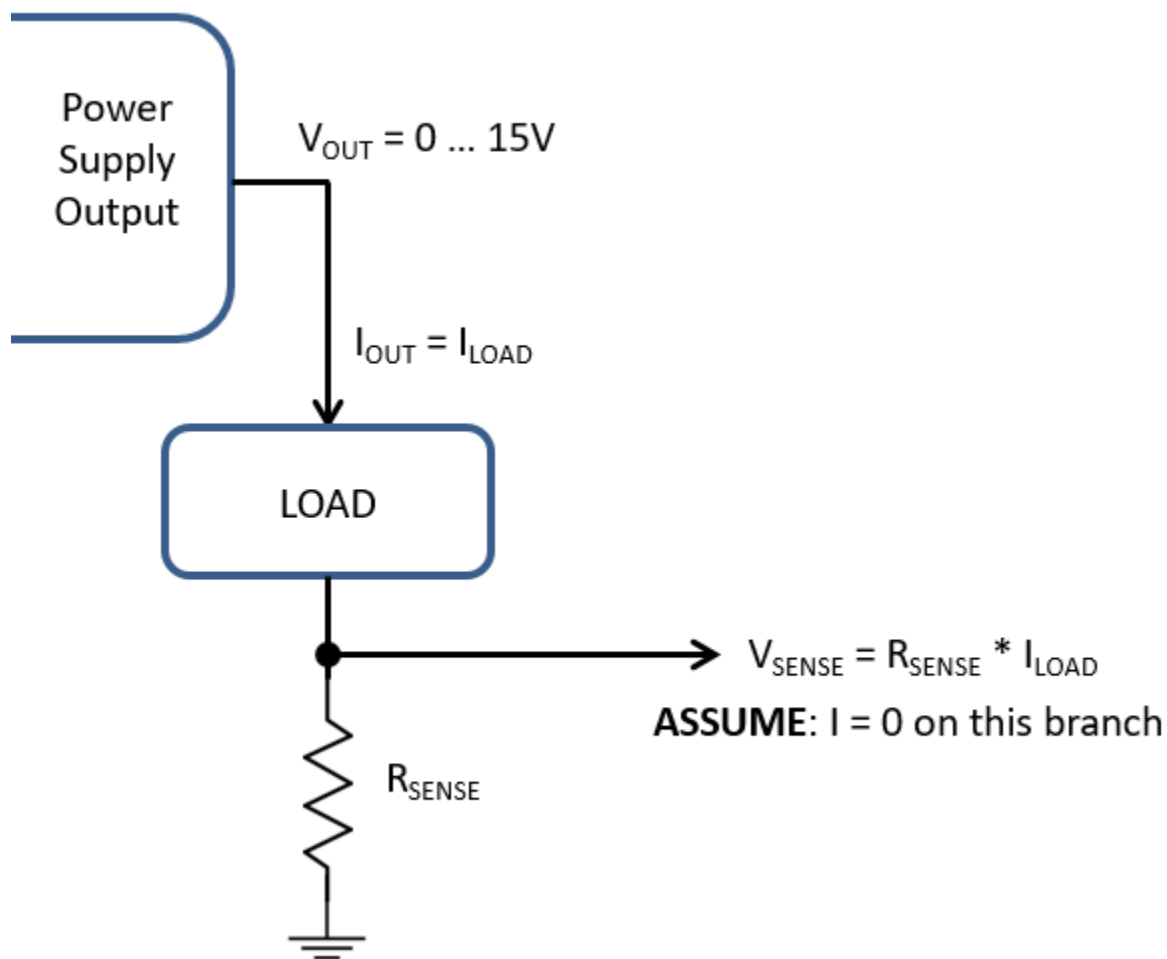


Figure 1 – Low-side current sensing

While it is tempting to use a large value for ease of measurement, such as $R_{SENSE} = 1\ \Omega$ (giving $1\ A = 1\ V$), this causes other problems. Any voltage drop across the resistor means the load will see that much less voltage (remember: the load sees a voltage of $V_{OUT} - V_{SENSE}$ in this arrangement) and any power dissipated in the resistor is wasted as heat (worsening our design's overall efficiency). Instead, we will try to minimize R_{SENSE} while still developing sufficient voltage to actually measure a usable voltage proportional to I_{LOAD} . We have selected $0.1\ \Omega$ as the lowest standard resistor value that is useful in this design – this is a *design constraint* for you. While lower standard resistor values do exist, going smaller will mean that other elements (such as circuit traces and wires) may dominate the resistance in the current sense path. If this is the case, the relationship between I_{LOAD} and the sensed voltage will be strongly subject to manufacturing and construction differences and other non-controlled variables. Noise will also become a problem as signal levels get smaller with decreased sensing resistance.

The location for R_{SENSE} we are using is called **low-side current sensing**. In general, current output can be measured in two places. One is on the **high side** – in the path connecting the power supply output to the top of the load. We can see from the diagram why this location is called *high-side*; the nominal voltage at this node will be equal to our power supply's full output (i.e., up to $+15V$). While this is a feasible location offering certain advantages, measuring the voltage differential across a resistor with up to $15V$ on one side and only slightly less on the other makes for a difficult circuit design. The following article (notably Page 2) discusses the limitations and benefits of each approach:

Linear Technology: Application Note 105, Current Sense Circuit Collection

<https://www.analog.com/media/en/technical-documentation/application-notes/an105fa.pdf>

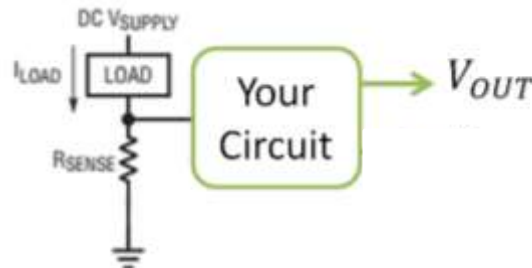
For our purposes, low-side sensing will function well enough and is much simpler to design – consider this a second *design constraint* for you. The major drawback, where an alternate path to ground allows current through the load to go un-sensed, is unlikely to occur in the expected use-cases for our design. We will have to expend slightly more design effort later to account for the voltage drop across R_{SENSE} . At high load currents, the ‘negative’ output terminal of our supply (lower connection to “LOAD” on the prior diagram) will not be at exactly the same potential as the $0\ V$ circuit ground, which slightly complicates later elements of the design.

YOUR Design Goals

We can now define the exact circuit we wish *you* to design as the deliverable for this assignment:

INPUT

- Your circuit will take in a varying current, I_{LOAD} , from 0 A to **3.5 A** returning from the load. This current should flow into a resistor, $R_{SENSE} = 0.1 \Omega$, which is to be part of your design. The low side of R_{SENSE} should be tied to ground. Overall, this should look like the following:



In the above, note the red-highlighted upper current limit. You may wonder if this is an error, as in the “background” document we decided on a 0 – 3A current-limited output for the overall supply. Sensing a load current slightly *higher* than 3 A is purposeful, however; in the case of a current *overshoot*, we need to be able to sense that the load current has actually *exceeded* the current limit. If we can only sense currents *up to* 3 A, then we might not be able correctly limit current at levels near 3 A, as our sensing circuit may ‘max out’. Allowing a small safety margin prevents this potential issue.

- When simulating your circuit in PSPICE, we recommend replacing the block ‘LOAD’ in the above with a DC current source (the “DC VSUPPLY” side can connect to any voltage, or ground). You can then use a “DC Sweep” simulation to sweep this source from 0 to 3.5A. Alternatively, you can replace the sense resistor *and* current source with a DC voltage source that sweeps from 0V to 0.35V.

OUTPUT

- Your circuit will have one output: a voltage ranging from 0.25 V to 3.3V, which is *proportional* to the current I_{LOAD} . In other words, if $I_{LOAD} = 0$ A, then $V_{OUT_1} = 0.25$ V, and if $I_{LOAD} = 3.5$ A, then $V_{OUT_1} = 3.3$ V.

CONSTRAINTS

- You may use *up to* 3 op-amps in your design (3 of the 4 contained in one LM324 op-amp chip). They will be of type LM324N (datasheet provided in assignment posting, see labs for LTSPICE model). These op-amps should be powered by a **single** supply voltage, $V_{SUPPLY} = +19V$ (that is, $V_+ = VCC = +19V$ and $V_- = VEE = 0V$).
 - NOTE:** Using fewer than three op-amps will make Design Assignment 2 easier – in this future assignment you will create a **filter** circuit with the left-over op-amps. Having more than one spare op-amp will allow you to create a better filter.
- You may use as many resistors as you wish. The current sense resistor, R_{SENSE} , must be 0.1Ω . All other resistors should be in the range of 1 K Ω to 100 K Ω , and should be selected as standard 5% values (see table in Appendix A for a list of standard values). Do not use series or parallel combinations to create total resistances outside of this range, though.
- You may *not* use any inductors or capacitors, or other components (diodes, transistors, etc.).

Deliverable

To complete Design Assignment 1, you must submit the following:

1. Engineering Specification and Background Research (< 1 page)

- Research and summarize potential building-block circuit topologies or elements relevant to this design. Identify core functionality, limitations, and a plan for integration.

2. Candidate Designs (< 2 pages)

- Create two *candidate designs* which address the problem at hand. We recommend that both group members collaborate closely on both designs rather than working individually. A complete candidate design consists of (i) a schematic or block diagram showing the solution, (ii) a *brief* description of functionality, possibly backed by selected design equations or other key results. Note that candidates do not necessarily have to be purposely divergent from one another, nor do they have to be fully designed circuits (block diagrams are fine). It is encouraged to take a promising design and iterate meaningfully on it from one candidate to the next, abstracting away circuit/component-level details where possible. However, if you find that your first design is not feasible, consider investigating a more fundamentally different design for the second candidate.

3. Final Candidate (< 3 pages)

- Based on lessons learned from the two candidate options, synthesize your final design from some combination of the two (and/or other elements – we just want to see iteration/evolution of the design). Summarize the result, including (i) an updated, fully-designed circuit schematic (if any modifications have been made since the original candidates), and (ii) a description of any such changes or modifications. As noted, you do not strictly have to select one candidate or the other. For instance, you may choose to combine the designs or otherwise modify them as your design process progresses.
- You should briefly discuss your final design and justify how you selected it based on the candidates provided.
- The final schematic should show all external connections: the $+19\text{ V}$ op-amp supply, the current input (I_{LOAD}), and the output (V_{OUT}).
- *You must simulate your final design under the conditions specified in the 'YOUR Design Goals' section (using LTSpice or other software). Your report must include a set of simulation results and brief discussion which uses these simulations to clearly demonstrate that your design is feasible and solves the problem at hand.*

Deliverable Constraints

The document is to be submitted online through the provided submission link. Please also consider the following:

- The document should be no more than *6 pages* with 11-pt minimum font size. Please submit in *PDF only*.
- A cover page should be included with the name and student number for both you and your partner.
- Extraneous sections such as an abstract, executive summary, table of contents, problem introduction / explanation, conclusion, etc. should be omitted. Please keep your overall report concise. It should be mostly schematics, design equations, and minimal supporting/explanatory text.
- The cover page and any appendices do not count towards the page limit. Use appendices for large graphics such as schematics and simulation results. Label figures and ensure they are legible/of appropriate size.
- Any use or application of existing works, including but not limited to circuits or circuit fragments, must be properly referenced in a clearly labelled reference section, with specific *numbered* references in the text.

Design Hints – Real-world Op-Amps and Voltage Levels

- You may have noticed that the output range for V_{OUT} seems to be selected very specifically. The reason behind this particular output range has to do with how the microcontroller (STM32G051) will ‘read’ the output we produce and how real-world op-amps operate. The chip has an on-board *analog-to-digital* converter (ADC); an analog voltage in the range of $0 \dots 3.3 \text{ V}$ applied to one of the analog input pins can be sampled as a 12-bit digital value (i.e., $0 \dots 4095$). Initially, one might think to have the μC read the $0 \dots 0.35 \text{ V}$ developed across R_{SENSE} directly (this voltage range is given by $R_{SENSE} = 0.1 \Omega$ and $I_{LOAD} = 0 \dots 3.5 \text{ A}$). This would mean that we use only $\frac{0.35 \text{ V}}{3.3 \text{ V}} = 10.6\%$ of the total range, or around 434 out of 4096 possible digital values. Each digital value would thus span a current range of $\frac{3500 \text{ mA}}{434} = 8 \text{ mA}$, meaning our sensing of the current would be somewhat coarse, especially at low current limit settings ($<100\text{mA}$). We would also be very susceptible to any noise in the signal or error introduced in the conversion process. Because of this it would be beneficial to **amplify** the $0 \dots 0.35 \text{ V}$ range to span the full $0 \dots 3.3 \text{ V}$ range of the ADC.
- It now appears we should output $V_{OUT} = 0 \dots 3.3\text{V}$ from our circuit. However, we expect that your design will use at least one op-amp to achieve its functionality. These op-amps will be powered by a **single** positive supply (not a dual supply). In particular, we will use the pre-regulated 19V from the laptop adapter to power this circuit. Thus, for an ideal op-amp, the possible output voltage range is $V_{Low \text{ Supply}} \dots V_{High \text{ Supply}}$, or $0 \dots 19\text{V}$ in our design. The problem which occurs is that real-world op-amps cannot set their output voltage all the way to the supply rails. Rather, V_{OUT} for any op-amp will take on a restricted range less than its two supply limits. For the LM324, one can see this limit in the provided data sheet, under “Output Voltage Low/High”:

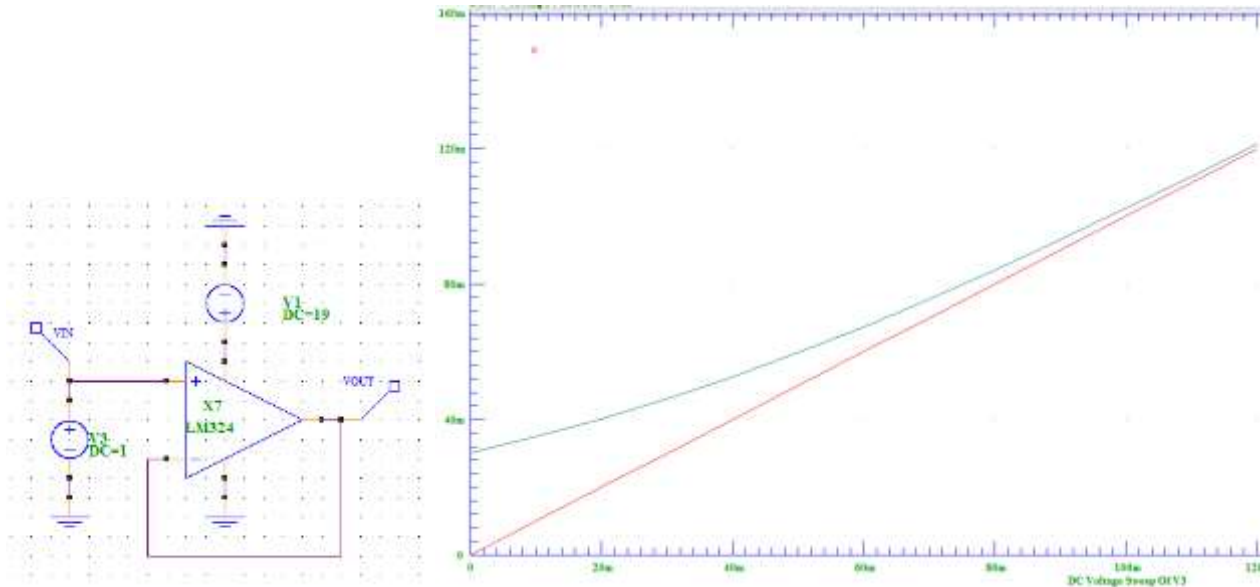
Output Voltage Swing	V_{OH}	$V^+ = 30 \text{ V (LM2902-N, } V^+ = 26 \text{ V)}$	$R_L = 2 \text{ k}\Omega$	22	V
			$R_L = 10 \text{ k}\Omega$	23 24	
	V_{OL}	$V^+ = 5 \text{ V, } R_L = 10 \text{ k}\Omega$		5 100	mV

- If this op-amp is powered with a 30V supply for instance, the datasheet tells us the output can go only from 0.1 V to 22 V under worst-case conditions (a resistive load attached to V_{OUT} , of value $2 \text{ k}\Omega$). Its behaviour is similar when powered with $+19 \text{ V}$ instead of $+30 \text{ V}$. While the upper limit (22 versus 30) is not an issue in this design (3.3V is far from 19V), the lower limit is. If we designed for a circuit output of $0 \dots 3.3\text{V}$, the circuit would not function correctly at low sensed currents. Since the op-amp’s output cannot go below $\sim 100\text{mV}$, we would always have $V_{OUT} \geq 0.10 \text{ V}$, even if $I_{SENSE} = 0 \text{ A}$. In other words, the output would not be correctly proportional to input current at low values. To avoid this we **shift** the range of our output voltage signal to avoid this lower limit by a safety margin, $0.25 \dots 3.3\text{V}$.
 - This tells us two key functions that your circuit must implement: **amplify** the input voltage and **shift** it (add an offset).

Common Single-Supply Mistakes

There are other implications of the prior discussion about real-world op-amp limits:

- Although we focused on the final output range, the voltage range limits imposed by the op-amp supply voltage apply to **all** op-amps in your design. In other words, if **any** op-amp in your design tries to output *any* voltage from $0V$ to $0.1V$, the *actual* output will be just $0.1V$. For example, here is a buffer circuit simulated with $V_{IN} = 0$ to $0.1V$:

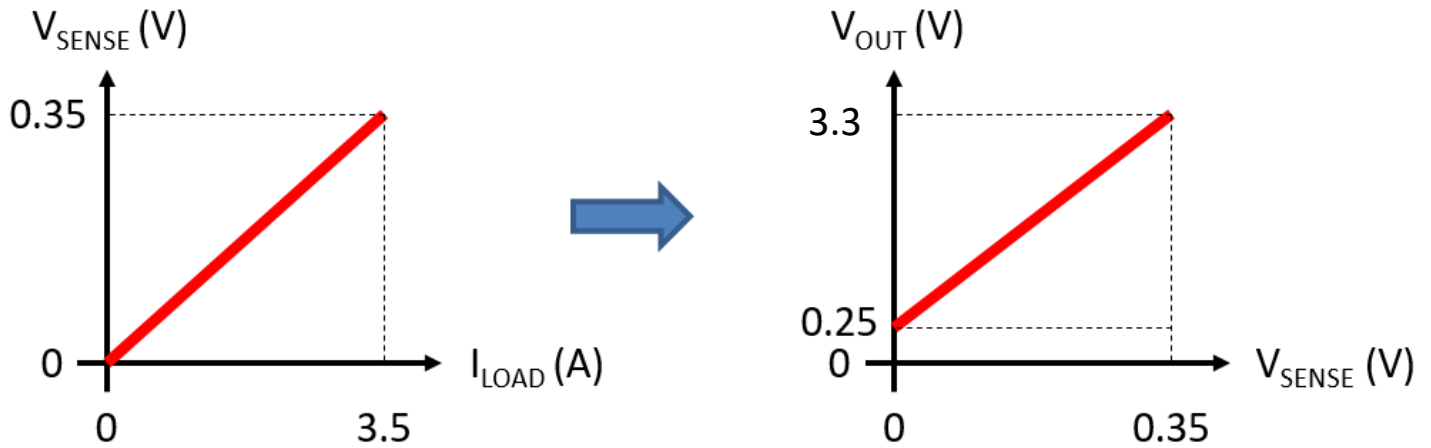


We expect to see $V_{OUT} = V_{IN}$ for all voltages, but instead we see that V_{OUT} (blue line) is increasingly different from V_{IN} (red line) the closer to zero V_{IN} gets. We generally model this (mis)behavior by simply saying that V_{OUT} for the op-amp “bottoms out” around $0.1V$ as per the datasheet.

- By extension, since voltages cannot go **below** zero volts (actually $0.1V$ as per above) due to the supply limits, your design cannot be expected to produce a **negative** voltage *anywhere* in the circuit. For this reason, be very cautious about using *inverting* amplifier circuits – a positive voltage into a basic inverting amplifier will produce a negative output (ideally), which in this real circuit will simply be close to zero volts. Note that if you absolutely need to use an inverting amplifier it is still feasible to do so, with some difficulty – you can research a “virtual ground” variation. Otherwise, consider favoring non-inverting configurations in your design.

Correct Operation

Correct operation of your circuit is achieved if you can demonstrate the following:



Some example points:

- If $I_{LOAD} = 0$, then $V_{SENSE} = (0.1)(0) = 0\text{ V}$, and V_{OUT} should be $V_{OUT} = \left(\frac{3.3-0.25}{0.35}\right)(0) + 0.25 = 0.25\text{ V}$.
- If $I_{LOAD} = 100\text{ mA}$, then $V_{SENSE} = (0.1)(0.1) = 0.01\text{ V}$, and $V_{OUT} = \left(\frac{3.3-0.25}{0.35}\right)(0.01) + 0.25 = 0.337\text{ V}$
- If $I_{LOAD} = 1\text{ A}$, then $V_{SENSE} = (0.1)(1) = 0.1\text{ V}$, and $V_{OUT} = \left(\frac{3.3-0.25}{0.35}\right)(0.1) + 0.25 = 1.121\text{ V}$
- If $I_{LOAD} = 3.5\text{ A}$, then $V_{SENSE} = 0.35\text{ V}$, and $V_{OUT} = \left(\frac{3.3-0.25}{0.35}\right)(0.35) + 0.25 = 3.3\text{ V}$

Design Hints – Circuit Toolkit

In prior design courses, you may have learned to organize fragments of problem solutions them into a toolkit for future use. In this course, we recommend doing the same with the circuits you analyze. We expect your incoming knowledge to include op-amp circuits from prior courses (and we will be reviewing several of these circuits in class). You can organize these into a basic toolkit of building blocks; the circuits you have access to include:

- *Basic amplifier* (inverting and non-inverting) – Multiplies a signal by a constant gain.
- *Summing amplifier* (inverting and non-inverting) – Adds two voltages together. Can also apply an individual weight to each voltage added by varying resistor values for each input branch.
- *Difference amplifier* (inverting and non-inverting) – Subtracts two input voltages. Can also apply a constant gain to the difference.
- *Buffer* (inverting and non-inverting) – Copies a signal input-to-output (possibly inverting it), and can be used to remove loading effects.
- *Two-Resistor Divider* – Can be used with the given $+19\text{ V}$ supply to create **any fixed voltage** in between. Be cautious of loading effects; for accuracy and precision we would recommend including an op-amp buffer if you are using a resistor divider to create an offset/constant voltage in your circuit.

You may of course make use of circuits not listed above. Also note that you may modify any of the above circuits in known ways:

- Op-amp feedback paths can be modified by adding additional parallel or series components to achieve new effects. One can even combine **multiple functions** into one op-amp circuit.
- Voltages can be ‘mixed’ at any node using a resistor in each incoming path (examine the summing amplifier designs for ideas).

We expect that your design can be readily completed using only the above building blocks, all of which you have seen in prior courses. However, you are free to look at other resources and use circuit designs not in the above list. You must still keep in mind that components other than the op-amps and resistors may not be used.

Design Hints – Starting Point

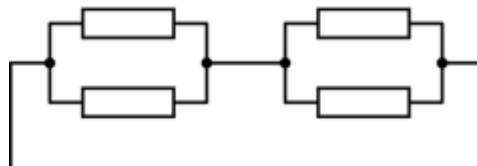
If you are having trouble deciding where to start, we recommend that you approach the problem in a manner similar to design problems in prior design courses. Your initial goal should be to apply insight to the specification and expand on its details as necessary, and to research existing similar problems and solutions. You are trying to break the larger, global task/function of your circuit into an unordered list of smaller sub-tasks or discrete steps which map well to the tool kit of building blocks you have available.

In other words, rather than approaching the problem as one intertwined task, think of it as a series of small modifications (add, multiply, etc.) to the signal coming into your circuit, eventually ending with the desired output. Once you have this list of steps, the problem becomes simpler: order the steps and add/subtract/merge steps as needed to ‘optimize’ the design. In this stage it can help immensely to create a block diagram of the task ordering before starting the actual circuit design.

Design Hints – General Best Practices and Miscellaneous Tips

In no particular order:

- If your input voltage (proportional to I_{LOAD} , $0 \dots 0.35 \text{ V}$ with $I_{LOAD} = 0 \dots 3.5 \text{ A}$ and $R_{SENSE} = 0.1 \Omega$) is graphed as a straight line (following the suggestion to create a current ‘ramp’ using a current source), the output voltages of your circuit should also be straight lines with no bends or curves if things are working correctly.
 - Zoom in particularly close to the origin to check the low-current input range, as any curvature here can be hidden when zoomed out.
- You will likely not be able to get the output range to be *exactly* $0.25 \dots 3.3 \text{ V}$ due to the rounding error associated with using standard resistor values. Exceeding or undershooting the 3.3 V limit by a few 10 ’s of mV is fine, but the lower limit has less of a safety factor ($250 - 100 = 150 \text{ mV}$). A larger lower limit (e.g., 300 mV versus 250 mV) is probably better in this context than an undershoot (e.g., 200 mV or less).
 - You generally shouldn’t need to do this to a significant extent, but if you find that an ideal resistor value is simply not available you can consider using two resistors in series or parallel to achieve intermediate results.
- While simulating your design in SPICE, note how the output changes when modifying component values such as resistances. If the change in output is very large for a small change in component value (say multiple volts difference and/or a large change in slope for a few 10 ’s of ohms change in resistance) then the circuit is very sensitive to component tolerance, and is unlikely to function well, if at all, in the real-world. The resistors we will use have a 5% tolerance, meaning their actual value may be up to $\pm 5\%$ different from their nominal value due to manufacturing variance.
- Although the final voltage range is $0.25 \dots 3.3 \text{ V}$, you are free to have voltages outside this range at intermediate points within your circuit if the effect is beneficial to your design. Be sure to still respect the overall voltage limits of the op-amps.
- You can achieve gain < 1 using a resistive divider.
- Be mindful of loading effects if you use any two-resistor dividers.
- COMPONENT VALUES – You will need to round resistor values to the range and options listed and this does introduce some error. You may not perfectly get a range of 0.25 V to 3.3 V , and this is fine, within reason. There is no need to do something like this:



Putting a pair of resistors in series or in parallel is fine to get an off-decade value, but more than two is unnecessary. The resistors used have a tolerance of 5% , meaning their actual value can be up to $\pm 5\%$ from the nominal value listed, so there is an inherent limit on the precision you can achieve. For instance, if you put two resistors, a 10 K and a 1 K in series, each resistor may be up to $\pm 5\%$ from its nominal. In that case the actual series resistance could be anywhere between $9.5 + 0.95 = 10.45 \text{ K}\Omega$ to $10.5 + 1.05 = 11.55 \text{ K}\Omega$ (versus the expected series resistance of $10 + 1 = 11 \text{ K}\Omega$).

Appendix A – Resistor and Capacitor Standard Values

Standard Resistor Values ($\pm 5\%$)						
1.0	10	100	1.0K	10K	100K	1.0M
1.1	11	110	1.1K	11K	110K	1.1M
1.2	12	120	1.2K	12K	120K	1.2M
1.3	13	130	1.3K	13K	130K	1.3M
1.5	15	150	1.5K	15K	150K	1.5M
1.6	16	160	1.6K	16K	160K	1.6M
1.8	18	180	1.8K	18K	180K	1.8M
2.0	20	200	2.0K	20K	200K	2.0M
2.2	22	220	2.2K	22K	220K	2.2M
2.4	24	240	2.4K	24K	240K	2.4M
2.7	27	270	2.7K	27K	270K	2.7M
3.0	30	300	3.0K	30K	300K	3.0M
3.3	33	330	3.3K	33K	330K	3.3M
3.6	36	360	3.6K	36K	360K	3.6M
3.9	39	390	3.9K	39K	390K	3.9M
4.3	43	430	4.3K	43K	430K	4.3M
4.7	47	470	4.7K	47K	470K	4.7M
5.1	51	510	5.1K	51K	510K	5.1M
5.6	56	560	5.6K	56K	560K	5.6M
6.2	62	620	6.2K	62K	620K	6.2M
6.8	68	680	6.8K	68K	680K	6.8M
7.5	75	750	7.5K	75K	750K	7.5M
8.2	82	820	8.2K	82K	820K	8.2M
9.1	91	910	9.1K	91K	910K	9.1M